Jobsubmit User Manual

Version 1.0.5

GNU General Public License (GPL)

Author

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1 Introduction

Jobsubmit is a lightweight, platform-independent job execution utility designed for scientific computing, batch processing, and reproducible workflows. It executes commands from a list file sequentially while clearly reporting job start time, completion status, execution duration, process ID, and exit code.

The design philosophy emphasizes simplicity, transparency, and deterministic behavior, making Jobsubmit suitable for local machines, login nodes, virtual machines, continuous integration pipelines, and development environments.

Jobsubmit is not intended to replace full-featured workload managers or schedulers. Instead, it provides a minimal and predictable execution layer for controlled, sequential workflows.

2 Key Features

Jobsubmit provides the following core features:

- Sequential execution of commands from a text file
- Explicit reporting of:
  - Job start date and time
  - Job completion or failure date and time
  - Execution duration in HH:MM:SS format
  - Process ID (PID)
  - Exit code
- Separate capture of standard output and standard error
- Clean, scientific-style console output

- Cross-platform compatibility (Linux, macOS, Windows)
- No external Python dependencies
- Simple installation and clean project layout

3 Project Structure

The Jobsubmit distribution follows a simple and transparent directory layout:

```
bin
   jobsubmit
docs
   jobsubmit_manual.tex
logs
tests
   cmd.lst
tools
   install.sh
.gitlab-ci.yml
.travis.yml
CHANGELOG
LICENSE
README.md
```

4 System Requirements

4.1 Software Requirements

- Python version 3.7 or newer
- Bash shell (required for the installer on Linux and macOS)

No third-party Python modules are required.

4.2 Hardware Requirements

Jobsubmit has no specific hardware requirements and can be executed on desktops, laptops, virtual machines, and CI runners.

5 Installation

5.1 Interactive Installation (Linux / macOS)

From the project root directory, execute:

```
bash tools/install.sh
```

The installer performs the following actions:

- Prompts for an installation directory
- Copies the Jobsubmit executable
- Sets executable permissions
- Optionally adds the installation directory to the user's PATH
- Creates output directories for standard output and error

After installation, reload the shell configuration:

```
source ~/.bashrc
```

5.2 Manual Installation

```
mkdir -p ~/jobsubmit
cp bin/jobsubmit ~/jobsubmit/
chmod +x ~/jobsubmit/jobsubmit
export PATH=$PATH:~/jobsubmit
```

To make the PATH modification permanent, add the export command to `~/.bashrc` or `~/.profile`.

5.3 Windows Installation

On Windows systems, Jobsubmit can be executed directly using Python:

```
python bin\jobsubmit cmd.lst all
```

Executable binaries generated using tools such as Nuitka or PyInstaller may also be used if available.

6 Usage

6.1 Command Syntax

```
jobsubmit <command_file> <line_selection>
```

6.2 Examples

```
jobsubmit cmd.lst all
jobsubmit cmd.lst 3
jobsubmit cmd.lst 1-5
jobsubmit cmd.lst 1,3,7
```

7 Command File Format

The command file (commonly named `cmd.lst`) defines the workflows to be executed.

- One command per line
- Blank lines are ignored
- Commands are executed sequentially

7.1 Example

```
echo "Job 1"
sleep 2
invalid_command
echo "Job 4"
```

8 Output Files

For each executed command, Jobsubmit generates:

- `stdout/<cmdfile>_line_<N>.out`
- `stderr/<cmdfile>_line_<N>.err`

These files enable post-run inspection, debugging, and archival.

9 Exit Codes

- Exit code `0`: Command executed successfully
- Non-zero exit code: Command execution failed

Jobsubmit itself exits with a non-zero status only in the event of fatal errors such as invalid input or missing files.

10 Continuous Integration

Jobsubmit includes configuration files for:

- GitLab CI (`.gitlab-ci.yml`)
- Travis CI (`.travis.yml`)

These configurations ensure consistent behavior across supported platforms.

11 License

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See the `LICENSE` file for the complete license text.

12 Author

Rajesh Prashanth Anandavadivel

Originally developed as a practical utility for managing reproducible scientific and computational workflows.

13 Changelog

All notable changes, version updates, and fixes are documented in the `CHANGELOG` file.

14 Philosophy

Jobsubmit is intentionally:

- Minimal

- Predictable
- Transparent
- Script-friendly

It is not intended to replace full-fledged schedulers such as SLURM or PBS, but to provide a reliable execution tool for controlled, sequential workflows.

15 Contributing

Bug reports, improvements, and suggestions are welcome. Contributions should remain focused, readable, and aligned with the scientific intent of the project.

End of User Manual